

Introduction to Calculus

Calculus is the mathematical study of continuous change. It has two major branches: differential calculus and integral calculus.

Derivatives

The derivative of a function measures the rate of change. For a function $f(x)$:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Common derivative rules:

- Power rule: $\frac{d}{dx}(x^n) = nx^{(n-1)}$
- Product rule: $\frac{d}{dx}(fg) = f'g + fg'$
- Chain rule: $\frac{d}{dx}(f(g(x))) = f'(g(x)) * g'(x)$

Applications include finding areas, volumes, and solving differential equations.